

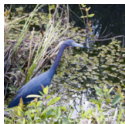
Invariance vs robustness

What orders a model's adversarial robustness

A project growing out of the practical course on adversarial robustness and shift invariance. Prior work relates the two and reaches opposite conclusions; we build a margin-based diagnostic and test what tells them apart, from small classifiers to foundation-model encoders.

The two properties, on one image

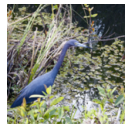
Adversarial robustness: a tiny worst-case change flips the label



clean: "little blue heron" ✓ ✓



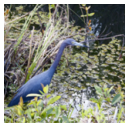
perturbation ($\times 10$)



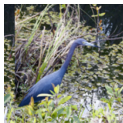
adversarial: "tusker" ✗

The two images differ by an ℓ_∞ change of $1/255$, invisible to a human; the label flips.

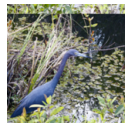
Invariance: shifting the image keeps the label



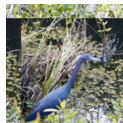
"little blue heron" ✓
original



"little blue heron" ✓
shift (20,0)



"little blue heron" ✓
shift (0,20)



"little blue heron" ✓
shift (28,28)

A real image and an ImageNet classifier. An ℓ_∞ change of $1/255$, invisible to a person, changes the label (top). Shifting the image does not (bottom).

The literature relates them, and disagrees

Invariance helps

- Zhang, ICML 2019. Anti-aliased pooling raises shift-consistency and corruption robustness.
- Saha & Gokhale, WACV 2024. “Better shift invariance is generally correlated with better adversarial robustness” (§6.4); PGD/FGSM, one ResNet-34.
- Grabinski et al., 2023. Alias-free downsampling improves robustness; FGSM/PGD/APGD.
- Wang et al., 2025. Equivariant CNNs improve robustness on CIFAR; FGSM and PGD.

Invariance hurts, or trades off

- Ge et al., NeurIPS 2021. A shift-invariant linear classifier uses only the input average; the margin drops from 2 to $2/\sqrt{d}$. PGD on MNIST/Fashion.
- Kamath et al., NeurIPS 2021. A proven trade-off between spatial and adversarial robustness; their rate of invariance is our consistency; rotation as a cyclic shift.

These lines use different data, models, and attacks. None on the left uses a parameter-free strong attack. Ge’s own reading: invariance hurts only when it both raises the dimension and shrinks the margin.

How we frame the project

What the project does

- defines a margin-to-sensitivity ratio and proves when it brackets the robust radius;
- shows shift-consistency does not order robustness and this ratio does, across model families;
- shows the “invariance helps” effect is a weak-attack artifact;
- adds the gradient’s shape to rank already-robust models.

Structure: (1) where the project started; (2) the ratio and its theory; (3) experiments; (4) ranking robust models; (5) limitations and questions.

What it does not claim

- that the ratio is a new object; it is a Lipschitz–margin certificate we refine;
- co-existence of invariance and robustness as a discovery;
- a new attack or a new defense; the ratio is a diagnostic.

Where this started

A recap of the practical course, in two points

- **What we did: reproduce and extend Ge et al. on MNIST and Fashion-MNIST.**

- Reproduced their ordering with FGSM and PGD (ℓ_2 , ℓ_∞): more shift-consistency went with lower robust accuracy across their models.
- Extended it to standard architecture choices, pooling type, kernel size, depth, an added dense layer, measuring robust accuracy and shift-consistency for each.

- **What we found, and what it left open.**

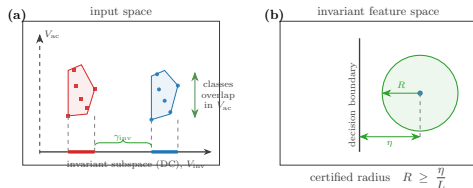
- The simple “more consistency, less robust” ordering broke: several CNNs were both shift-consistent and as robust as the fully-connected baseline. We used a consistency-versus-robustness graph to screen architectures.
- It did not say which quantity predicts robustness when consistency does not, and it used only weak attacks. That is what the rest of the project addresses.

From the project to now

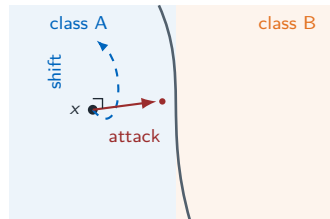
Recent papers, and what is still open

- **The co-existence we saw is now in a published paper.** During the practical course we found that invariance and robustness can co-exist. TIPS ([Saha & Gokhale, 2024](#)) later reported the same correlation, so this observation is now established in the literature.
- **There are still additions we can make on top of it.** A parameter-free strong attack was never applied to this line; the invariance metric was never tested for whether it *orders* robustness; and there is no quantity that predicts robustness without an attack.
- **The question.** Is there a quantity, computable without an attack, that orders adversarial robustness across models, where shift-consistency does not?

Invariance and robustness are different directions



Imposing invariance projects the classes onto the invariant part (their average); robustness is the distance $R \geq \eta/L$ to the boundary.



The shift walks *around* the boundary (label kept); the attack walks *toward* it (label crossed). Near-orthogonal directions.

Observation

For margin $M(x) = f_y(x) - \max_{j \neq y} f_j(x)$, a perturbation δ moves the margin by at most $\|\nabla_x M(x)\|_q \|\delta\|_p$. Hence the model keeps its label while $\|\delta\|_p < M(x) / \|\nabla_x M(x)\|_q$ (Lipschitz–margin certificate, [Hein 2017](#); [Tsuzuku 2018](#)).

- **Threat-matched.** We match the dual norm to the attack: $q=1$ for ℓ_∞ , $q=2$ for ℓ_2 . The mismatched norm gives the wrong sign (shown later on MNIST).
- **Scale-free.** Rescaling logits $f \mapsto cf$ scales M and $\|\nabla M\|$ together, so the ratio is unchanged: robustness attaches to the ratio, not to margin or Lipschitz alone.
- **Per model:** $\eta/L = \eta/L$, the mean margin over the mean dual-norm gradient on clean, correctly classified images. No attack is needed to compute it.

Proposition 1: invariance fixes the usable margin

Proposition 1 (Invariant linear margin)

A linear score is shift-invariant iff its weight is constant, $w = c\mathbf{1}$. For any finite separable classes X_+, X_- the invariant ℓ_2 max-margin is

$$\gamma_{\text{inv}} = \frac{1}{2} \text{dist}(\text{conv } \Pi_{\text{inv}} X_+, \text{conv } \Pi_{\text{inv}} X_-),$$

the separation of the class averages, with direction $\pm\mathbf{1}/\sqrt{d}$.

In words: a shift-invariant linear classifier can only use the average of the input, so the only margin it has is the gap between the class averages, which is small. This is Ge's $2/\sqrt{d}$ collapse; we prove it for any dataset, not only the single-image case.

Proposition 2: the ratio brackets the true radius

Proposition 2 (Two-sided bracket)

If the score is L -Lipschitz toward the boundary and α -co-Lipschitz along a path reaching it, then

$$\frac{\eta}{L} \leq r \leq \frac{\eta}{\alpha}, \quad 1 \leq \frac{r}{\eta/L} \leq \kappa := \frac{L}{\alpha}.$$

The bracket is exact ($\kappa = 1$) for linear invariant features.

In words: η/L is not just a lower bound; a matching upper bound holds, so the true radius is trapped between them. The two ends are close, $\kappa \in [1.6, 2.4]$ in our tests, so η/L is essentially the radius. The lower bound is Hein/Tsuzuku; the upper bound is what we add.

Does shift-consistency order robustness? Does the ratio?

Capacity-matched dissection: same parameters, only the invariance operator differs

Four operators (standard, anti-aliased, exactly cyclic, shift-augmented), identical parameter counts. Data: CIFAR-10, MNIST, Fashion; small CNNs and a PreActResNet-18. Radius by a minimum-norm attack; robustness under adversarial training by AutoAttack (Croce & Hein 2020), gradient-masking checked.

We find that shift-consistency does not order robustness, and anti-predicts it under adversarial training, while the threat-matched η/L orders it.

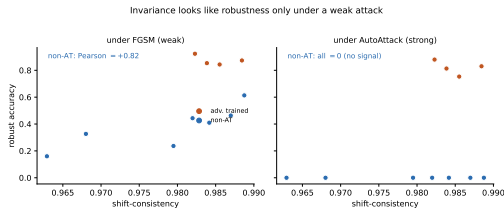
	η/L	consistency
CIFAR-10 (standard)	+0.998	−0.33 (n.s.)
ResNet-18 (adv. tr.)	+0.91	−0.81
ImageNet-100 (adv. tr.)	+0.90	−0.87

Correlation with the robust radius (standard) or AutoAttack (adversarial). η/L predicts beyond clean accuracy (partial +0.89 on the ResNet cells).

Is the “invariance helps” effect real?

Repeat it under a weak attack and a strong one

- We reproduce the “helps” regime (standard training, FGSM/PGD) on our operators and on those of [Saha 2024](#) and [Wang 2025](#), then add AutoAttack.
- Under FGSM the ordering appears; under AutoAttack every non-robust model drops to near zero and it disappears. On frozen CLIP encoders, FGSM reports 16–61% that AutoAttack erases.

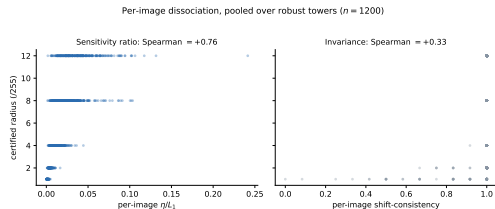


Same encoders and budget: consistency looks predictive under FGSM (left), predicts nothing under AutoAttack (right).

We find the “invariance helps” correlation is a weak-attack artifact; no “helps” paper had applied a parameter-free strong attack.

Per image, where training cannot confound

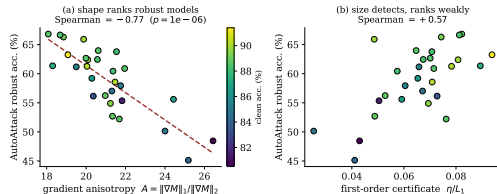
- On sixteen frozen CLIP encoders shift-consistency is *saturated*, 0.963–0.988, robust or not, so it cannot rank them.
- Within one robust encoder, image by image, η/L orders each image's radius at Spearman +0.74 to +0.96; per-image consistency from +0.37 down to about zero. Inside one encoder, training is not the confound.



Pooled over robust encoders ($n=1200$): η/L (left) orders the per-image radius; consistency (right) does not.

Ranking already-robust models: the gradient's shape

- Among robust models η/L no longer orders them. Add the gradient's anisotropy $A = \|\nabla M\|_1 / \|\nabla M\|_2 \in [1, \sqrt{d}]$: an ℓ_∞ attack wastes budget on a concentrated gradient, exploits a spread one.
- Over thirty RobustBench CIFAR-10 models, lower A is more robust: Spearman -0.79 (-0.77 controlling clean accuracy). Replicates on MNIST/Fashion.
- **Proposed mechanism** (not a theorem): a spread-over-dimensions deficit $r \approx (\eta/L_1)/(1 + CA)$, checked on a controlled model and thirty real ones.



30 RobustBench models. (a) the shape A ranks robustness;
(b) the size η/L detects but ranks weakly.

The results across settings

setting	attack / metric	η/L	consist.	shape A
CIFAR-10 (standard)	min-norm radius	+0.998	−0.33 n.s.	—
MNIST / Fashion (AT sweep)	AutoAttack ℓ_∞	predicts	n.s.	−0.61 to −0.89
PreActResNet-18 (adv. tr.)	AutoAttack ℓ_∞	+0.91	−0.81	—
ImageNet-100 (adv. tr.)	AutoAttack ℓ_∞	+0.90	−0.87	—
16 CLIP encoders (per image)	min-norm radius	+0.74 to +0.96	≈ 0	—
30 RobustBench models	AutoAttack ℓ_∞	detects	—	−0.77

Pearson or Spearman as appropriate; η/L is threat-matched (dual norm matched to the attack). **n.s.** = not statistically significant; ≈ 0 = does not order the radius; — = not applicable or not measured. The shape A is computed only where we *rank already-robust* models (RobustBench, and the MNIST/Fashion training-strength sweep), so it is blank in the other rows; shift-consistency was not measured for the downloaded RobustBench models. Every number reproduces from the released code and result files.

Where it breaks

- **The ratio is not a trainable target.** It is scale-free, so its direct maximizer is the constant classifier. It is a diagnostic, not a loss.
- **The shape law needs range.** On low-dimensional data the anisotropy is compressed and does not rank until a training-strength sweep widens the range; the clean cross-model evidence is RobustBench.
- **The deficit is proposed, not proven.** We verify it on a controlled model and thirty real ones, but it is not a network-level theorem.
- **Text-modality robustness stays narrow.** On CLIP the worst case over a small paraphrase set is too mild to form a second adversarial axis.

Proposed contributions, and existing results

Verified against the primary sources

Proposed contribution	Existing result we use or refine
Consistency does not order robustness while threat-matched η/L does, across model families	The certificate $r \geq M/\ \nabla M\ $: Hein & Andriushchenko 2017 ; Tsuzuku et al. 2018
The powered per-image dissociation within one frozen encoder (training cannot confound)	Attack-free radius estimates: CLEVER (Weng et al. 2018)
The “invariance helps” effect vanishes under a parameter-free strong attack	That invariance and robustness are related: Ge ; Kamath ; Saha ; Wang
The upper arm of the bracket, $r \leq \eta/\alpha$, with an explicit condition number κ	The DC-collapse / margin mechanism: Ge et al. 2021
Anisotropy $\ \nabla M\ _1/\ \nabla M\ _2$ ranks already-robust models	Certified radii via smoothing: Cohen et al. 2019
The effective-dimension deficit as its (proposed) mechanism	Gradient-dimension and curvature views: Simon-Gabriel 2019 ; Moosavi 2019
Saturation of shift-consistency on modern encoders, and its textual analogue	Participation-ratio of the input gradient (overfitting detection): 2025
The margin-sensitivity sign-flip under adversarial training	AutoAttack as the arbiter: Croce & Hein 2020 ; margin consistency in robust models: Ngnawe 2024

- Shift-consistency, the metric this literature tracks, does not order adversarial robustness, and the “invariance helps” effect vanishes under a strong attack.
- A threat-matched, scale-free margin-to-Lipschitz ratio, which we complete to a two-sided bracket, orders robustness from small classifiers to frozen CLIP encoders.
- To rank already-robust models, the shape of the gradient (its anisotropy) matters, through a spread-over-dimensions deficit.

Next directions: beyond ℓ_∞ ; the rich-regime optimization / implicit-bias question; a genuine text-adversarial axis; scaling the anisotropy study beyond RobustBench.

Where your input would help most

- **Positioning as a certificate.** η/L is a Lipschitz–margin certificate we complete to a two-sided bracket. Given your work on robustness certificates, is the bracket a meaningful addition, or is it subsumed by existing certified methods (randomized smoothing, Lipschitz certification)? What is the cleanest framing of “certificate vs diagnostic”?
- **The kernel / lazy-regime theory.** Our lazy-regime result treats orbit-averaging as a projection in the tangent-kernel space. Given your deep-net-to-kernel-machine work, is this the right lens, and where are the pitfalls (e.g. the bare-ReLU NTK is not Lipschitz)?
- **Weakest parts / experiments.** Which pieces look weakest to you, the small robust-encoder panel, the range-dependence of the anisotropy law, the proposed-not-proven deficit, and which experiments would you rerun or design differently?
- **Literature and framing.** Are we missing work at the certified-robustness $\times$ invariance intersection? And is the dissociation, or the anisotropy law, the stronger headline?